

Simulations on Infinite Size Lattices

H.G. Evertz, W. von der Linden

Institut für Theoretische Physik, Technische Universität Graz, 8010 Graz, Austria
(August 4, 2000)

We introduce a Monte Carlo method, as a modification of existing cluster algorithms, which allows simulations directly on systems of infinite size, and for quantum models also at $\beta = \infty$. All two-point functions can be obtained, including dynamical information. When the number of iterations is increased, correlation functions at larger distances become available. Limits $q \rightarrow 0$ and $\omega \rightarrow 0$ can be approached directly. As examples we calculate spectra for the d=2 Ising model and for Heisenberg quantum spin ladders with 2 and 4 legs.

Standard Monte Carlo simulations are limited to systems of finite size. Physical results for infinite systems have to be obtained by finite size scaling, assuming that one knows the correct scaling laws, and assuming that the data are already in a suitable asymptotic regime. It is therefore very desirable to obtain results also *directly* at infinite system size. We will show how to do so, with only a small modification of existing cluster algorithms, by using the cluster representation of the models to calculate two-point functions. In the quantum case we can then also simulate directly at $\beta = \infty$ and calculate correlation functions and dynamical greens functions. As examples we will show calculations for the classical Ising model and for quantum Heisenberg ladder systems.

The *Swendsen Wang cluster method* [1] for the classical Ising model is based on the Fortuin-Kasteleyn representation

$$e^{\beta J(s_i s_j - 1)} = \sum_{b_{ij}=0,1} p \delta_{s_i s_j} \delta_{b_{ij},1} + (1-p) \delta_{b_{ij},0} \quad (1)$$

$$=: \sum_{b_{ij}=0,1} W_{ij}(s_i, s_j, b_{ij}) \quad (2)$$

for the Boltzmann weight of a spin-pair, with $p = 1 - e^{-2\beta J}$, which enlargens the phase space of spin variables s_i by additional bond variables b_{ij} . The partition function $Z = \sum_{\{s_i\}} \sum_{\{b_{ij}\}} W(s, b)$ with total weight $W(s, b) = \prod_{\langle ij \rangle} W_{ij}(s_i, s_j, b_{ij})$ is then simulated efficiently by switching between representations: given a spin-configuration $s := \{s_i\}$, one generates a bond configuration $b := \{b_{ij}\}$ with the conditional probability $p(b|s) \sim W(s, b)$, thus creating a configuration of *clusters* of sites connected by bonds $b_{ij} = 1$. Given a bond configuration, a new spin configuration is generated with probability $p(s|b) \sim W(s, b)$. Because of the factor $\delta_{s_i s_j} \delta_{b_{ij},1}$ in W_{ij} , this amounts to setting all spins of each cluster randomly to a common new value, independent of other clusters.

Observables $\hat{O}$ can be computed either in spin-representation as $O(s)$ or in bond representation as $O(b) = (\sum_s O(s) W(s, b)) / (\sum_s W(s, b))$ (so called “improved estimators”) [2]. For two-point functions

$O(s) = s_i s_j$ the bond representation is particularly simple:

$$O(b) = \delta(\text{sites } i \text{ and } j \text{ are in the same cluster}) . \quad (3)$$

Thus two-point functions and derived quantities (including susceptibility, energy, specific heat) can be computed from the properties of individual clusters.

A variant of the Swendsen-Wang method is Wolff’s *single cluster method* [2]. Given a spin configuration, only one of the bond-clusters is constructed, namely the cluster which contains a randomly chosen initial starting site x_0 . All spins in this cluster are then flipped. One advantage is that the single cluster will on average be larger than the average Swendsen-Wang cluster (so that its flip results in a bigger move in phase space). Correcting for this fact, one obtains [2]

$$\langle s_i s_j \rangle = \langle \frac{V}{V_{cl}} \delta(i \text{ and } j \text{ in the single cluster}) \rangle , \quad (4)$$

where V_{cl} is the number of sites in the single cluster, and the brackets $\langle \dots \rangle$ on the right refer to the Monte Carlo average.

Our new *infinite system method* now employs the single cluster method, except that it starts each cluster not from a randomly chose site, but always from the *same* site x_0 (e.g. the origin of the coordinate system). For the ferromagnetic Ising model, we begin with an initial staggered spin configuration of unlimited size. (Only a finite part if this configuration will have to be stored). We iterate the following two steps: (1) For the current spin configuration, a cluster containing site x_0 is constructed with Swendsen-Wang bond-probabilities, and (2) all spins in this cluster are flipped, resulting in a new spin configuration. The current cluster is discarded after each iteration.

After a sufficient total number $N(r)$ of iterations, this process will “equilibrate” all spins within a radius r around x_0 , and two-point functions can then be measured within this area. With increasing number of iterations, the cluster will occasionally reach larger and larger distances. Because of eq. (4), the probability to do so is (roughly) proportional to the correlation function $C(r)$. A region of the lattice at distance r from x_0 will become

“equilibrated” after the cluster has reached that region a sufficient number n_{eq} of times, thus $N(r) \sim \frac{n_{eq}}{C(r)}$. Since the cluster can grow without bounds, we are calculating correlation functions for the *infinite* lattice, while we only need to store the spin configuration within the area actually reached by a cluster during a finite run.

Fig. 1 shows as an example the correlation function for the two dimensional Ising model at $\beta = 0.42 < \beta_c = 0.44068\dots$. For comparison we show results for finite size

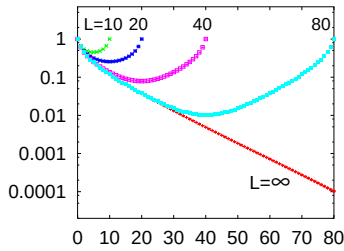


FIG. 1. Correlation functions of the $d = 2$ Ising model at $\beta = 0.42$.

lattices, with periodic boundary conditions. Usually, the infinite lattice result can only be obtained as the extrapolation of the finite lattice ones. In contrast, with our new method we can obtain the infinite lattice result in a single simulation, producing the lower straight line in fig. 1. The computational effort per sweep is about the same as for the $L = 80$ system. We checked that the method works just as well in three dimensions.

The calculation of error bars for $C(r)$ needs special care. For each distance $|r|$, we demand that the cluster contains pairs of sites at that distance $|r|^{d-1} \times n_{eq}$ times, i.e. it reaches each site at that distance $O(n_{eq})$ ($= O(10)$) times, before we consider data at that distance thermalized. Only then do we start to accumulate measurements for $C(r)$ according to eq. (4) for all sites i, j in the cluster. Alternative similar procedures are possible.

When will the new method work? The main constraint is that the computational effort, proportional to the average cluster size, must be finite. There must not be a finite probability for a percolating cluster, thus we need $\beta < \beta_c$. If we assume $C(r) \sim e^{-r/\xi}/r^{d-2+\eta}$, then the average cluster size is finite for either $\xi < \infty$, or for $\xi = \infty$ and η sufficiently large. The present implementation of the new method will thus work if the unsubtracted correlation function decays sufficiently rapidly. Systems with long range order may become accessible with a suitable modification of the method.

For *nonrelativistic quantum systems*, the loop algorithm [3] provides a cluster method. It is based on an enlarged representation in terms of the original spin operators and additional loop operators, similar in spirit to the Fortuin-Kasteleyn representation, eq. (1). We will in the following specialize to the spin $\frac{1}{2}$ quantum Heisenberg model

$$\mathcal{H} = J \sum_{\langle ij \rangle} \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) + \lambda S_i^z S_j^z. \quad (5)$$

The two-point function for the single cluster version of the loop algorithm then reads

$$\begin{aligned} \left\langle \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) \right\rangle &= \left\langle \frac{V}{V_{cl}} \delta(\text{i and j on the loop}) \right\rangle, \\ \langle S_i^z S_j^z \rangle &= \left\langle S_i^z S_j^z \frac{V}{V_{cl}} \delta(\text{i and j on the loop}) \right\rangle, \end{aligned} \quad (6)$$

where V_{cl} is the number of sites on the loop. Therefore our approach to simulate infinite size systems can be used in the same way as for the Ising model. “Infinite size” here can be applied to the spatial directions, as well as, independently, to the direction of imaginary time, yielding simulations directly at $\beta = \infty$, while retaining all dynamical information. Note that our method is not a projector method. We can simulate directly at $L = \infty$ and $\beta = \infty$. This also enables us to obtain the limits $q \rightarrow 0$ and $\omega \rightarrow 0$ directly from the simulations. Contributions from $q \equiv 0$ or $\omega \equiv 0$, which are in general different from those limits, and are present in normal simulations with periodic boundary conditions, are completely avoided in our approach, as well as the double limit $L \rightarrow \infty$ and $q \rightarrow 0$ that is usually required. When the number of iterations increases, clusters will reach larger distances in spatial and/or imaginary time direction, thereby improving the resolution in q and ω .

Again, our present method will be applicable provided that the unsubtracted correlation function drops off sufficiently quickly (for which finite correlation length and/or finite gap are sufficient but not necessary).

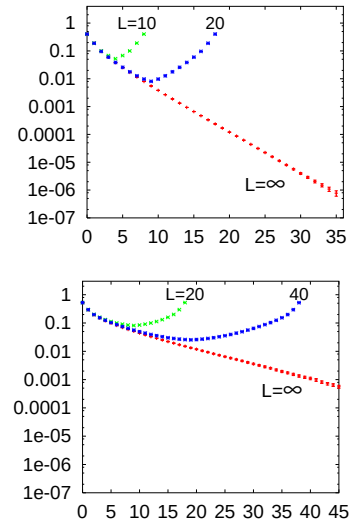


FIG. 2. Equal time staggered spatial correlation function at $q_{\perp} = \pi$ for $N=2$ (top) and $N=4$ (bottom)

As an example we studied spin ladder systems [4] with $N = 2$ and 4 legs for the isotropic antiferromagnet

($\lambda = 1$). We used the discrete time version of the loop algorithm ($\Delta\tau = 1/16$). The method can be applied just as well in continuous time [6]. Fig. 2 shows results for the equal time staggered spatial correlation functions along the chains, which behave similarly to the Ising case. A fit to the infinite lattice result gives $\xi = 2.93(2)$ for $N = 2$ and $\xi = 8.2(1)$ for $N = 4$. Fig. 3 shows greens functions

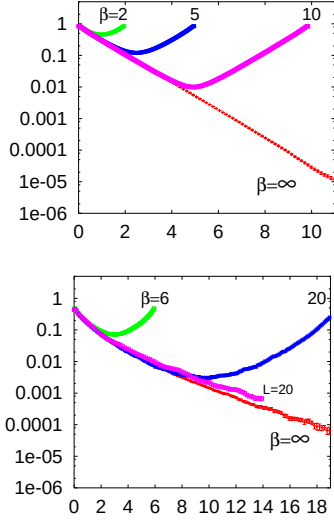


FIG. 3. Greens functions $\langle S(\vec{q}, 0) S(\vec{q}, t) \rangle$ at $\vec{q} = (\pi, \pi)$ for $L = \infty$, the infinite size system, with $N = 2$ (top) and $N = 4$ (bottom)

for $L = \infty$, the infinite size system. Whereas finite temperature calculations give results periodic in imaginary time, which have to be extrapolated, the new approach provides the $\beta = \infty$ ($T = 0$) result directly. With increased number of iterations, the infinite system results become available at larger distances both in spatial and in imaginary time direction. A fit to the exponential decay $G(\tau) \sim e^{-\tau\Delta}$ of our data directly provides estimates for the gaps $\Delta = 0.5059(4)$ at $N = 2$ and $\Delta = 0.19(1)$ at $N = 4$, consistent with previous results [5]. We also show results for $L = 20$ and $\beta = \infty$ to exemplify the effect of finite size systems.

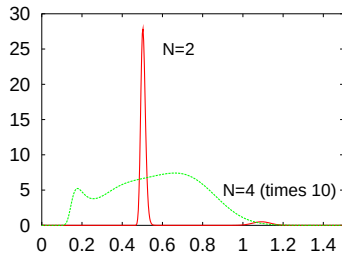


FIG. 4. Spectrum $S(\vec{q}, \omega)$ at $\vec{q} = (\pi, \pi)$ for $L = \infty$ and $\beta = \infty$

Continuing the imaginary time greens function to real

frequencies by the maximum entropy method provides the spectra of fig. 4, in which the gaps, the single magnon peaks, and higher excitations for $N = 4$ are clearly visible. This appears to be the first time that the spectrum for $N = 4$ has been calculated.

Let us compare our method to other approaches. Exact diagonalization provides ground state results but is limited to small systems. Projector methods are also limited in system size and rely critically on proper convergence to the ground state. DMRG can achieve $\beta = \infty$ (mostly without dynamical information) for fairly small systems, or infinite size for large temperatures and finite $\Delta\tau$. The most powerful method to extrapolate to infinite system size is the Finite Size Scaling method of Kim [7] and Caracciolo et al. [8], which allows extrapolation at correlation lengths far larger than the system size, but requires knowledge about scaling and corrections to scaling of the model.

In summary, we have introduced a new method, as a small modification of existing cluster methods, to simulate both classical and quantum systems at infinite system size and/or at zero temperature, while obtaining all two-point functions. Larger distances (in space and/or imaginary time) are accessed for longer simulations. The method in its present form is applicable as long as the unsubtracted two-point correlation function decays sufficiently quickly.

-
- [1] R.H. Swendsen and J.S. Wang, Phys. Rev. Lett. 58, 86 (1987) ; for a review see e.g. A.D. Sokal, *Bosonic Algorithms*, in M. Creutz, “Quantum Fields on the Computer” (Advanced Series on Directions in High Energy Physics, Vol 11); newer version available at <http://www.dbwilson.com/exact/cargese.ps.gz>.
 - [2] U. Wolff, Phys. Rev. Lett. 62, 361 (1989), Nucl. Phys. B 334, 581 (1990)
 - [3] H.G. Evertz, G. Lana, and M. Marcu, Phys. Rev. Lett. 70, 875 (1993), for a review see H.G. Evertz, cond-mat/9707221 (2nd ed. Jan. 2000)
 - [4] See e.g. E. Dagotto and M. Rice, Science 271, G18 (1996)
 - [5] B. Frischmuth, B. Ammon, and M. Troyer, Phys. Rev. B 54, R3714 (1996); S.R. White, R.M. Noack, and D.J. Scalapino, Phys. Rev. Lett. 73, 886 (1994)
 - [6] B.B. Beard and U.J. Wiese, Phys. Rev. Lett. 77, 5130 (1996)
 - [7] J.K. Kim, Phys. Rev. Lett. 70, 1735 (1993) , Phys. Rev. D 50, 4663 (1994)
 - [8] S. Caracciolo et al., Phys. Rev. Lett. 74, 2969 (1994) , Phys. Rev. Lett. 75, 1891 (1995)